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Sub ject : DSA

Unit 2, S14, SLO2 - Applications - Sparse Matrix, Polynomial Arithmetic, Josephus Problem

1. In a sparse matrix, most of the elements are __.
 Answer: zero (0)

2. A linked list is preferred for representing sparse matrices to save __.
 Answer: memory (space)

3. Each node in the linked list representation of a sparse matrix typically stores row index, column index, and the __.
 Answer: value (non-zero element)

4. Using linked lists for sparse matrices allows efficient __ of non-zero elements.
 Answer: storage and manipulation

5. Compared to a 2D array, a linked list avoids storing multiple __ elements in a sparse matrix.
 Answer: zero

6. In the linked■list representation of a polynomial, each node typically contains a coefficient and an __.
 Answer: exponent (power)

7. To add two polynomials represented as linked lists, you traverse both lists simultaneously and merge nodes with the same __.
 Answer: exponent (power)

8. In polynomial multiplication using linked lists, for each term in the first list, you multiply it by every term in the second list and insert the result into a __ linked list.
 Answer: result (new)

9. When inserting a new term into the result list during multiplication, if a node with the same exponent already exists, you simply add to its __.
 Answer: coefficient

10. Keeping the linked list of polynomial terms in descending order of exponents simplifies operations like addition, subtraction, and __.
 Answer: multiplication / evaluation

11. The Josephus problem is a theoretical problem related to a certain elimination game in a __.
 Answer: circle

12. A __ linked list is often used to simulate the Josephus problem efficiently.
 Answer: circular

13. In the Josephus problem, counting proceeds around the circle and every __ person is eliminated.
 Answer: k-th (m-th)

14. Using a circular linked list, we can efficiently move from one node to the __ during elimination.
 Answer: next

15. The last remaining person in the Josephus problem is called the __.

Answer: survivor / leader